Memory Based Q & A

1. Which university introduced Expert systems?

A. Massachusetts Institute of Technology
B. University of Oxford
C. Stanford University
D. University of Cambridge

Ans: C

Explanation: Expert System introduced by the researchers at Stanford University, Computer Science Department.

2. What is the function of an Artificial Intelligence "Agent"?

A. Mapping of goal sequence to an action
B. Work without the direct interference of the people
C. Mapping of precept sequence to an action
D. Mapping of environment sequence to an action

Ans: C

Explanation: A math function that converts a collection of perceptions into actions is known as the agent function. The function is implemented using agent software. An agent is responsible for the actions performed by the machine once it senses the environment.

3. Consider the traversal of a tree:
Preorder →ABCEIFJDGHKL
Inorder→ EICFJBGDKHLA

A. EIFJCKGLHDBA
B. FCGKLHDBUAE
C. FCGKLHDBAEIJ
D. IEJFCGKLHDBA

Ans: D

4. X=0110 and Y=11001 are two 5 bit binary numbers represented in two's complement format. The sum of X and Y represented in two's complement format using 6 bits is

A. 110111
B. 001000
C. 000111
D. 101001

Ans: C

5. Which bus is responsible for transferring data?

A. Address Bus
B. Control Bus
C. Data Bus
D. None of the above

Ans: C

6. A French curve is used to draw

A. Circle
B. Ellipse
C. Smooth free form curve
D. Polygon

Ans: C

7. Which one of the following is not a reduction scale?

A. 1:1 B. 1:200
C. 5/320 D. 5:2

Ans: A

8. For drawing of small instruments, watches etc. the scale used is

A. Reduced scale B. Full scale
C. Enlarged scale D. None of these

Ans: A

9. When the drawing are drawn smaller than the actual size of object then scale is known as

A. Reduced scale
B. Enlarged scale
C. Full scale
D. None of these

Ans: A

10. If the 10m length is represented as 1 mm on the map then representative fraction is

A. 1/100 B. 1/1000
C. 1/10 D. None of these

Ans: D

11. Which of the following is not the right number system?

A. Hexadecimal B. Binary
C. Roman D. Octal

Ans: C

12. **The Y-shear preserves the.................. coordinates and changes the ....coordinates.**
A. X, X B. Y, Y
C. X, Y D. Y, X

Ans: C

13. **The maximum power is delivered from a source to its load when the load resistance is ______ the source resistance.**
A. greater than
B. less than
C. equal to
D. less than or equal to

Ans: C

14. **Which instruction is used for return from service routine?**
A. IRET B. RETI
C. RET D. INTR

Ans: B

15. **Which of the following is the loop?**
A. For B. While
C. Do While D. All of the above

Ans: D

16. **In Colpitt's oscillator, feedback is obtained....................**
A. By magnetic induction
B. By a tickler coil
C. From the center of split capacitors
D. None of the above

Ans: C

17. **....................is a fixed frequency oscillator**
A. Phase-shift oscillator
B. Hartely-oscillator19
C. Colpitt's oscillator
D. Crystal oscillator

Ans: D

18. **How is a J-K flip flop made to toggle?**
A. When J=1 and K=1
B. When J=1, K=0
C. When J=0, K=1
D. None of the above

Ans: A

19. **How many pins are there in 8255?**
A. 40 Pins B. 45 Pins
C. 55 Pins D. 60 Pins

Ans: A

20. **What is the default port number for HTTP?**
A. 80 B. 8080
C. 443 D. 21

Ans: A

21. **In an AC system, what is the relationship between the peak voltage (V_{peak}) and and the root the root mean square voltage (V_{rms})?**
A. $V_{Peak} = V_{rms}$
B. $V_{peak} = 2 \times V_{rms}$
C. $V_{peak} = \frac{1}{\sqrt{2}} \times V_{rms}$
D. $V_{peak} = \sqrt{2} \times V_{rms}$

Ans: C

22. **In a Class A amplifier, what can be said about energy dissipation?**
A. Energy dissipation is minimal because the amplifier operates efficiently.
B. Energy dissipation is high as the amplifier conducts current throughout the entire input cycle.
C. Energy dissipation is zero since Class A amplifiers are known for complete energy conservation.
D. Energy dissipation is directly proportional to the square of the output power.

Ans: B

23. **What is the value of the maximum efficiency of the class B amplifier?**
A. 25%
B. 35%
C. 35% to 50%
D. 50% to 70%

Ans: D

Explanation: Class B amplifiers are more efficient compare to the class A amplifier because of good protection against noise effects.

24. How many transistors must be used in a class B power amplifier to obtain the output for the full cycle of the signal?

A. 0 B. 1
C. 2 D. 3

Ans: C

25. What is the default port number for HTTPS (Hypertext Transfer Protocol Secure)?

A. 80 B. 443
C. 8080 D. 21

Ans: C

26. What is the time complexity of Quick Sort in the average case?

A. O(n)
B. O(n log n)
C. O(n^2)
D. O(log n)

Ans: B

27. In Quick Sort, which element is chosen as the "pivot" during the partitioning step?

A. The first element in the array
B. The middle element in the array
C. A randomly chosen element from the array
D. The last element in the array

Ans: C

28. What is the main advantage of Quick Sort over other sorting algorithms like Bubble Sort and Insertion Sort?

A. Stable sorting
B. In-place sorting
C. Linear time complexity
D. Quadratic time complexity

Ans: B

29. What are the dimensions of an A4 size paper in inches?

A. 8.5 x 11
B. 8.27 x 11.69
C. 9 x 12
D. 8 x 10.5

Ans: B

30. Which ISO standard defines the A4 paper size?

A. ISO 9001 B. ISO 14001
C. ISO 216 D. ISO 27001

Ans: C

31. What does the term "AAU" stand for in the context of computer architecture?

A. Arithmetic and Analysis Unit
B. Address Allocation Unit
C. Arithmetic Address Unit
D. Advanced Arithmetic Unit

Ans: C

32. In a computer system, what role is typically associated with the Arithmetic Address Unit (AAU)?

A. Performing arithmetic calculations
B. Managing memory addresses
C. Handling input/output operations
D. Executing control instructions

Ans: B

33. Which component is responsible for translating logical addresses into physical addresses in computer systems?

A. CPU
B. ALU (Arithmetic Logic Unit)
C. MMU (Memory Management Unit)
D. GPU (Graphics Processing Unit)

Ans: C

34. How is generalization implemented in Object Oriented programming languages?

A. Inheritance B. Polymorphism
C. Encapsulation D. Abstract Classes

Ans: A

Explanation: None.

35. Which type of bus operates independently of a clock signal, allowing data transfers to occur without being synchronized to a fixed clock frequency?

A. Synchronous Bus
B. Clockless Bus
C. Serial Bus
D. Parallel Bus

Ans: B

36. An enhancement type NMOS transistor with $V_1 = 0.7$ V and source terminal grounded and 1.5 V applied to the gate. In what region does the device operate for $V_D = 0.5$ V?

A. Cut off region

B. Saturation region

C. Triode region

D. None of the above

Ans: C

37. Which of the following values of an alternating voltage or current represent the real magnitude ?

A. RMS value

B. Peak value

C. Average value

D. Instantaneous value

Ans: A

38. What are the four major elements of a use case diagram?

A. Actors, Classes, Associations, and Use Cases

B. Objects, Inheritance, Attributes, and Relationships

C. Actors, Systems, Relationships, and Use Cases

D. Modules, Functions, Variables, and Interfaces

Ans: C

39. Among the given options, which search algorithm requires less memory?

A. Optimal Search

B. Depth First Search

C. Breadth-First Search

D. Linear Search

Ans: B

Explanation: *The Depth Search Algorithm or DFS requires very little memory as it only stores the stack of nodes from the root node to the current node.*

40. Which of the given language is not commonly used for AI?

A. LISP B. PROLOG

C. Python D. Perl

Ans: D

Explanation: *Among the given languages, Perl is not commonly used for AI. LISP and PROLOG are the two languages that have been broadly used for AI innovation, and the most preferred language is Python for AI and Machine learning.*

41. Which one of the following refers to the copies of the same data (or information) occupying the memory space at multiple places.

A. Data Repository

B. Data Inconsistency

C. Data Mining

D. Data Redundancy

Ans: D

Explanation: The data redundancy generally occurs whenever more than one copy of the exact same data exists in several different places. Sometimes it may cause data inconsistency, which can result in an unreliable source of data or information that is not good for anyone.

42. Which of the following is a condition that causes deadlock?

A. Mutual exclusion

B. Hold and wait

C. Circular wait

D. No preemption

E. All of these

Ans: E

Explanation: None

43. A digital circuit that can store only one bit is a

A. Register B. NOR gate

C. Flip-flop D. XOR gate

Ans: C

44. Minimum number of flops required for modulus 15 counter is

A. 15 B. 16

C. 4 D. 3

Ans: C